

CSSE3113 – Introduction to Software Engineering

Requirements Engineering Process & Discovery Techniques

Requirements Engineering Process

Goal of RE Process → to create and maintain a system requirement document

- RE processes may Include four high level activities
 - Assessing whether the system is useful to business and customer (Feasibility Study)
 - Discovering requirements (elicitation and analysis)
 - Converting theses requirements into some standard form (specification)
 - Checking that the requirements actually define the system that the customer wants (validation)

Requirements Engineering Process

- The process used for RE vary widely depending on the **application domain**, the **people** involved and the **organization** developing the requirements.
- However, there are a number of generic activities common to all processes.
 - Requirements Elicitation
 - Requirements Analysis
 - Requirements Validation
 - Requirements Management

Feasibility Study

- A feasibility study decides whether or not the proposed system is worth implementing.
- A feasibility study is short, focused study that take place early in the RE Process
 - Does the system contributes to **organizational objectives**?
 - Can the system be implemented **within schedule and budget using current technology**?
 - Can the system can be **integrated** with other systems that are used?
- If the answer to any of these questions is no, you should probably not go ahead with the project.

Requirement Elicitation and Analysis

- Sometimes called requirement discovery.
- Involves technical staff (software engineers) working with customers to find out about the
 - application domain
 - the services that the system should provide and
 - the system's operational constraints.

May involve a variety of people in an organization

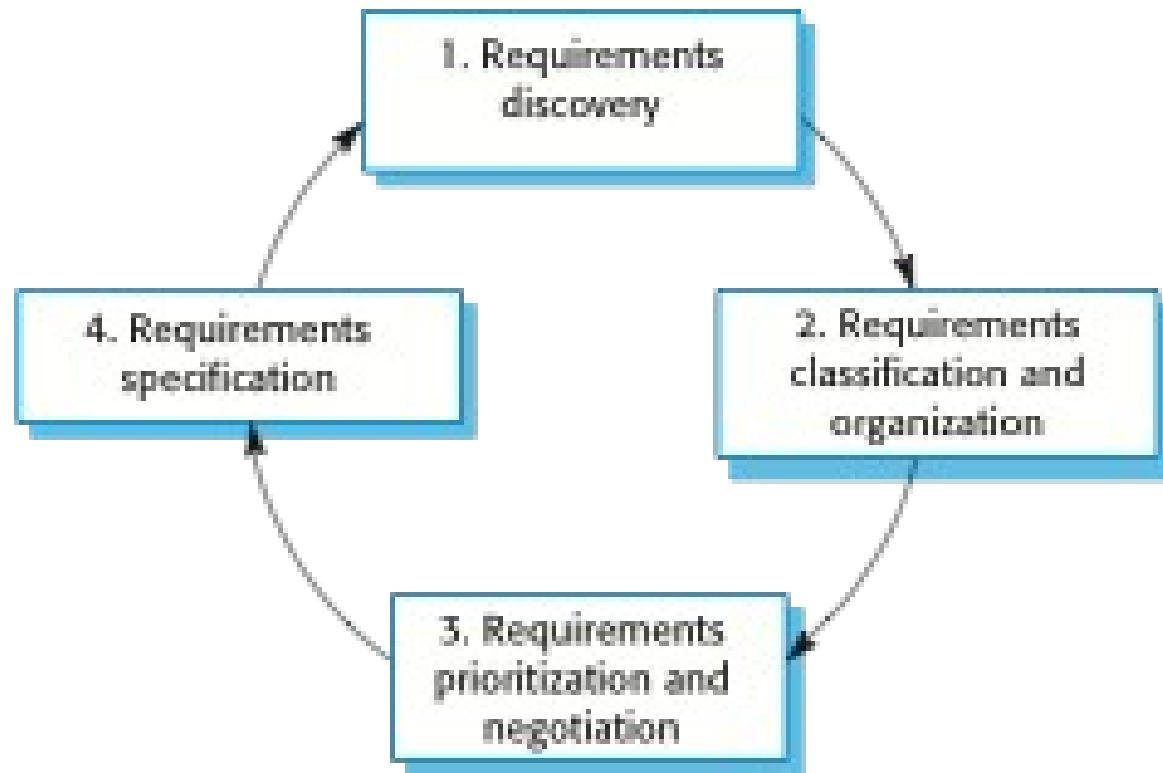
- End-users, managers, engineers involved in maintenance, domain experts etc. These are called *stakeholders*

Requirement Elicitation and Analysis

Process Activities

- Requirements discovery
 - Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.
- Requirements classification and organization
 - Groups related requirements and organizes them into coherent clusters.
- Prioritization and negotiation
 - Prioritizing requirements and resolving requirements conflicts through negotiation.
- Requirements documentation(Specification)
 - Requirements are documented.

Requirements Elicitation and Analysis Process



Problems of Requirements Analysis

Understanding stakeholder requirements is difficult for several reasons

- Stakeholders don't know what they really want.
- Stakeholders express requirements in their own terms.
- Different stakeholders may have conflicting requirements.
- Organizational and political factors may influence the system requirements.
- The requirements change during the analysis process. New stakeholders may emerge and the business environment change.

Requirements Discovery

- The process of gathering information about the proposed and existing systems
- Refining the user and system requirements from this information.
- Sources of information include system stakeholders and the specifications of similar systems.

Techniques of Requirement Discovery

Interview:

- Formal or informal interview with the system stakeholders
- In this, the RE team puts questions to stakeholders about the system that they currently use and the system to be developed.
- There are two types of interview
 - **Closed interviews:** where the stakeholders answers a pre-defined set of questions.
 - **Open interviews:** in which there is no pre-defined agenda and a range of issues are explored with stakeholders.

Effective Interviewers

- Interviewers should be open-minded.
- Avoid pre-conceived ideas about the requirements, and willing to listen to stakeholders.
- If the stakeholder comes up with surprising requirements , willing to change their minds about the system.
- They prompt the interviewee to get with discussion using a question or a proposal.

User Stories

- User stories are commonly used in adaptive or agile methodologies
- Short, high-level descriptions of required functionality expressed in customer terms.
- A typical user story has the form: “As a <role>, I want <goal/desire> so that <benefit>.”
- Just enough information so that the developers can produce a reasonable estimate of the effort to implement it.
- The aim is to avoid some of the waste that often happens in projects where detailed requirements are gathered early but become invalid before the work begins.

Example

Downloading and printing an article

First, you select the article that you want from a displayed list. You then have to tell the system how you will pay for it - this can either be through a subscription, through a company account or by credit card.

After this, you get a copyright form from the system to fill in and, when you have submitted this, the article you want is downloaded onto your computer.

You then choose a printer and a copy of the article is printed. You tell the system if printing has been successful.

If the article is a print-only article, you can't keep the PDF version so it is automatically deleted from your computer.

Scenarios

- People usually find it easier to relate to real-life examples rather than abstract descriptions.
- Scenarios are real-life examples of how a system can be used.
- Through Scenario user can understand and criticize....how they might interact with a software system.
- Requirement engineers can use the information gained from this discussion to formulate the actual system requirements.

Scenarios

They should include:

- A description of the starting situation
- A description of the normal flow of events
- A description of what can go wrong
- Information about other concurrent activities
- A description of the state when the scenario finishes.

LIBSYS Scenario

Initial assumption: The user has logged on to the LIBSYS system and has located the journal containing the copy of the article.

Normal: The user selects the article to be copied. He or she is then prompted by the system to either provide subscriber information for the journal or to indicate how they will pay for the article. Alternative payment methods are by credit card or by quoting an organisational account number.

The user is then asked to fill in a copyright form that maintains details of the transaction and they then submit this to the LIBSYS system.

The copyright form is checked and, if OK, the PDF version of the article is downloaded to the LIBSYS working area on the user's computer and the user is informed that it is available. The user is asked to select a printer and a copy of the article is printed. If the article has been flagged as 'print-only' it is deleted from the user's system once the user has confirmed that printing is complete.

LIBSYS Scenario

What can go wrong: The user may fail to fill in the copyright form correctly. In this case, the form should be re-presented to the user for correction. If the resubmitted form is still incorrect then the user's request for the article is rejected.

The payment may be rejected by the system. The user's request for the article is rejected.

The article download may fail. Retry until successful or the user terminates the session.

It may not be possible to print the article. If the article is not flagged as 'print-only' then it is held in the LIBSYS workspace. Otherwise, the article is deleted and the user's account credited with the cost of the article.

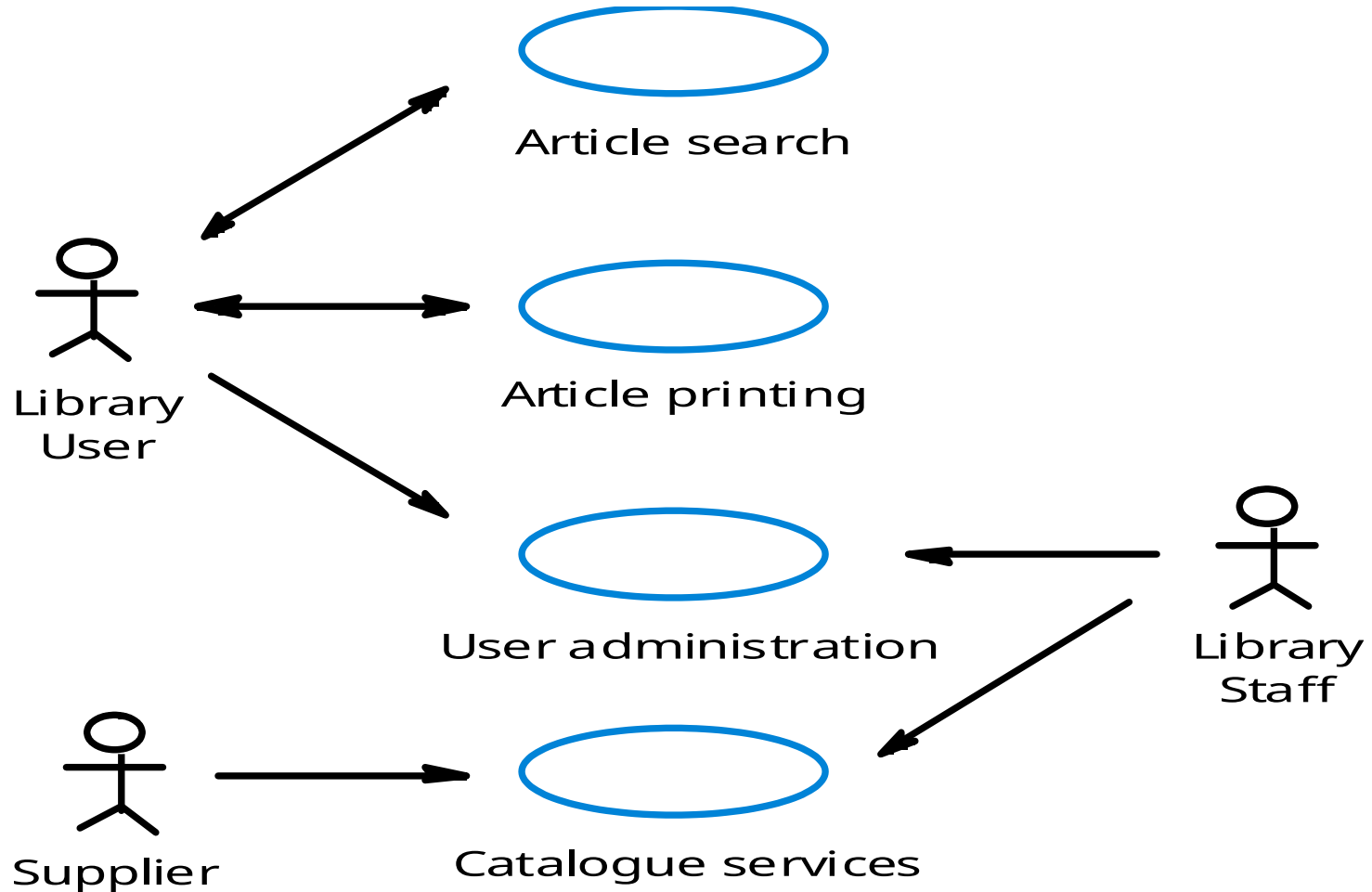
Other activities: Simultaneous downloads of other articles.

System state on completion: User is logged on. The downloaded article has been deleted from LIBSYS workspace if it has been flagged as print-only.

Use Cases

- Use-cases are a scenario based technique for requirement elicitation
- Which identify the actors in an interaction and which describe the interaction itself.
- A set of use cases should describe all possible interactions with the system.
- Sequence diagrams may be used to add detail to use-cases by showing the sequence of event processing in the system.

LIBSYS Use Cases



Social and Organizational Factors

- Software systems do not exist in isolation – they are used in a social and organizational context.
- This can influence or even dominate the system requirements.
- Good analysts must be sensitive to these factors.

Example

- Consider a system which allows senior management to access information without going through middle managers
 - Managerial status. Senior managers may feel that they are too important to use a keyboard. This may limit the type of system interface used
 - Managerial responsibilities. Managers may have no uninterrupted time where they can learn to use the system
 - Organisational resistance. Middle managers who will be made redundant may deliberately provide misleading or incomplete information so that the system will fail

Ethnography

- Ethnography is an observational technique that can be used to understand social and organizational requirements.
- An analyst immerses him or herself in the working environment (where the system will be used)
- Help analyst to discover implicit system requirements.

Requirements Validation

- Concerned with demonstrating that the requirements define the system that the customer really wants.
- Requirements error costs are high so validation is very important
- Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.

Requirements Checking

Checks include

- **Validity.** Does the system provide the functions which best support the customer's needs?
- **Consistency.** Are there any requirements conflicts?
- **Completeness.** Are all functions required by the customer included?
- **Realism.** Can the requirements be implemented given available budget and technology

Requirements Validation Techniques

- **Requirements reviews:**
 - Systematic manual analysis of the requirements (by a team of reviewers) who check for error and inconsistencies.
- **Prototyping:**
 - Using an executable model of the system to check requirements.
- **Test-case generation:**
 - Developing tests for requirements to check testability.

Requirements Management

- Requirements management is the process of **managing changing requirements** during the requirements engineering process and system development.
- New requirements emerge during the process as business needs change and a better understanding of the system is developed